Finite $SU(3)$ Lattice Gauge Theory from Particle Trajectories

Abstract

We construct a finite, explicit $SU(3)$ lattice gauge theory from the discrete trajectory data of a finite particle system. Starting from a recorded trajectory data set, we extract a finite lattice of sites together with temporal worldline links and same-time interaction links. At each site we form a complex color triplet by modulating the viscous force vector with a kinematic phase factor of de Broglie type; these triplets encode momentum direction in the internal color space. Link transport in $SU(3)$ is then defined from short-link color increments. Canonical plaquette holonomies, discrete field strengths, a finite Wilson action, and covariant fermionic and scalar matter actions are constructed on the same lattice. The complete data package $\mathcal{Q}$ is proved to be finite and $SU(3)$ gauge-invariant. In the $N \to \infty$ thermodynamic limit, scaled link and plaquette sums converge in probability to integrals against the deterministic stationary mean-field law π_∞. The limit theory is a bilocal gauge theory: the connection

$$A(z,z') := \frac{1}{2}\sum_{a=1}^{8} \mathrm{Re}\big(c(z)^\dagger \lambda_a \big(c(z') - c(z)\big)\big)\lambda_a, \qquad z, z' \sim \pi_\infty,$$

the parallel transport $U(z,z') := \exp(ig_3 A(z,z'))$, and the mean-field Wilson action

$$S_{\mathrm{W}}^{\infty} = \iiiint \Big(1 - \tfrac{1}{3}\,\mathrm{Re}\,\mathrm{tr}\, U(z_1,z_2)U(z_2,z_3)U(z_3,z_4)U(z_4,z_1)\Big)\, d\Gamma_{\Box}^{\infty}(z_1,z_2,z_3,z_4)$$

are all defined directly on the continuous mean-field measure, with no lattice re-discretization required. Path-integral quantization and axiom verification are deferred to companion papers.

Contents

1 Introduction

Lattice gauge theory provides a non-perturbative regularization of quantum field theories by replacing continuous spacetime with a discrete lattice and assigning $SU(N)$ group elements to the links. In the standard formulation the link variables are *independent integration variables* of a path integral, and the gauge group is imposed by hand. The present paper takes a different starting point: we ask whether a natural $SU(3)$ gauge structure can be *extracted* from discrete trajectory data, without imposing any gauge group a priori.

The data source is the finite-particle kinetic core studied in the companion convergence paper [1]. At each discrete timestep, each particle i experiences a force

$$\mathbf{F}_i^{\mathrm{visc}} = \nu_{\mathrm{visc}} \sum_{j \neq i} \omega_{ij}(v_j - v_i)$$

that pulls its velocity toward a weighted average of its interaction partners. This force is a vector in $\mathbb{R}^3$ and therefore lives naturally in the fundamental representation of $O(3)$.

The key observation is that $\mathbb{R}^3$ is three-dimensional, matching the defining dimension of the fundamental representation of $SU(3)$, the gauge group used throughout this paper. By complexifying the viscous force with a kinematic phase factor analogous to the de Broglie phase $e^{ip\ell/\hbar}$, each site acquires a complex color triplet $\mathbf{c}_x \in \mathbb{C}^3$. The color triplets at two neighboring sites define a short-link color increment from which we build an $SU(3)$ parallel transport $U_3(e)$ for every link e. No additional input is required: the gauge group, the link variables, and the coupling strength all emerge from the dynamics of the viscous particle system.

In detail, we first build a complex color triplet at each site, then project the short-link color increment onto the Gell-Mann basis to obtain a Hermitian traceless edge generator. The coupling parameter $g_3 > 0$ is placed to match the usual continuum normalization in

$$D_\mu = \partial_\mu - i g_3 A_\mu^a \lambda_a / 2.$$

We then add temporal worldline links and define canonical elementary plaquettes from one time step and one persistent interaction link. This restores the local 2-cell geometry needed for the mean-field Wilson action without reintroducing any extra edge nomenclature.

This paper supplies all the formal definitions and proofs required for a subsequent axiom-verification paper. The finite-data result is Theorem 8.1: the tuple

$$\mathcal{Q} = (\text{lattice}, \{U_3(e)\}, S_{\mathrm{W}}, S_{\mathrm{F}}, S_\phi)$$

is finite, explicit, and $SU(3)$ gauge-invariant. The thermodynamic graph and plaquette limits are unconditional consequences of the companion convergence paper together with the present graph construction. The mean-field result is stated in Section 9: under the $N \to \infty$ thermodynamic limit, the Wilson action converges to a bilocal integral functional defined directly on the stationary mean-field law.

Organization. Section 2 fixes notation, imports kernel parameters from the companion convergence paper, and tabulates all scalar parameters used throughout. Section 3 constructs the finite lattice $\mathcal{L} = (\mathcal{N}, \mathcal{E})$ from the recorded data, including worldline and interaction links. Section 4 defines the viscous force on sites and links and proves its covariance under orthogonal transformations. Section 5 introduces color triplets, the Gell-Mann basis, and the $SU(3)$ link transport. Section 6 defines path holonomies, canonical plaquettes, discrete field strengths, the Wilson action, and proves gauge invariance. Section 7 constructs covariant fermionic and scalar matter actions. Section 8 packages the complete data object and states the finiteness theorem. Sections 9 and 10 discuss the lattice-gauge interpretation, the thermodynamic graph and plaquette limits, the bilocal mean-field Yang–Mills action arising in the $N \to \infty$ regime, and quantization.

2 Notation and setup

Definition 2.1 (Kernel parameters and trajectory data)**.** *Let $d = 3$. The spatial dimension is fixed to three throughout because the viscous force is a vector in physical space and its three components form a natural triplet for the fundamental representation of $SU(3)$. Fix $N \geq 1$ and the index set $\mathcal{I} := \{1, \ldots, N\}$. At each time every trajectory entry is recorded for all $i \in \mathcal{I}$. The alive set $\mathcal{A}(t) \subseteq \mathcal{I}$ records which indices are active. Fix kernel parameters*

$$\ell_{\text{visc}} > 0, \qquad \kappa_0 > 0, \qquad \nu_{\text{visc}} \geq 0,$$

and define the regularized Gaussian kernel

$$K_\rho(x, y) := \kappa_0 + \exp\left(-\frac{\|x - y\|^2}{2\ell_{\text{visc}}^2}\right), \qquad \rho := \ell_{\text{visc}}.$$

The kernel satisfies

$$\kappa_* := \inf_{x,y} K_\rho(x, y) = \kappa_0 > 0, \qquad \kappa^* := \sup_{x,y} K_\rho(x, y) = 1 + \kappa_0 < \infty.$$

These bounds are imported from the companion paper on kinetic convergence; no other results from that paper are used here.

A recorded trajectory is available at discrete times

$$\mathbb{T} = \{0, 1, \ldots, T\}, \quad T \in \mathbb{N},$$

with finite alive sets $\mathcal{A}(t) \subseteq \mathcal{I}$ and position, velocity, and fitness data

$$x_i(t),\, v_i(t) \in \mathbb{R}^3, \qquad \Phi_i(t) \in \mathbb{R}, \qquad i \in \mathcal{I},\ t \in \mathbb{T}.$$

At each timestep t, the finite set $\mathcal{P}_t \subseteq \mathcal{A}(t) \times \mathcal{A}(t) \setminus \{(i, i) : i \in \mathcal{A}(t)\}$ records the ordered interaction companions used by the algorithm.

The following table summarizes all scalar parameters introduced in this paper, together with their role and the constraint they must satisfy.

Symbol	Role	Constraint
ℓ_{visc}	Viscous kernel length scale	> 0
κ_0	Kernel floor (regularization)	> 0
ν_{visc}	Viscous coupling strength	≥ 0
Δt	Physical time spacing between recorded slices	> 0
g_3	Gauge coupling constant	> 0
ℓ_0	Length unit for de Broglie phase	> 0
$\hbar_{\text{eff}}$	Effective action quantum	> 0
μ_p	Momentum coupling to phase	> 0
ϵ_c	Color regularization	> 0
η	Distance regularization in scalar action	> 0
m_F	Fermion mass	> 0
m_ϕ	Scalar witness mass	> 0

Lemma 2.2 (Kernel orthogonal symmetry)**.** *For every orthogonal matrix $O \in O(3)$ and all $x, y \in \mathbb{R}^3$,*

$$K_\rho(Ox, Oy) = K_\rho(x, y).$$

Proof. By definition,

$$\|Ox - Oy\|^2 = \|O(x - y)\|^2 = \|x - y\|^2,$$

so the Gaussian factor is invariant under simultaneous orthogonal transformation. □

Lemma 2.3 (Orthogonal covariance of viscous forces)**.** *Let $O \in O(3)$ and define transformed data $x_i'(t) = Ox_i(t)$, $v_i'(t) = Ov_i(t)$. Then for each site $x = n_{i,t}$,*

$$\mathbf{F}^{\text{visc}}(x; \textit{transformed data}) = O\, \mathbf{F}^{\text{visc}}(x; \textit{original data}).$$

Proof. Lemma 2.2 gives $K_\rho(Ox_i, Ox_j) = K_\rho(x_i, x_j)$, hence

$$\deg'_{\text{link}}(i, t) = \deg_{\text{link}}(i, t), \qquad \omega_{ij}^{\text{link}\prime}(t) = \omega_{ij}^{\text{link}}(t).$$

For each link $e = (n_{i,t} \to n_{j,t})$,

$$\mathbf{F}^{\text{visc}\prime}(e) = \nu_{\text{visc}}\, \omega_{ij}^{\text{link}}(t)(Ov_j - Ov_i) = O\, \mathbf{F}^{\text{visc}}(e).$$

Summing outgoing links at a fixed source site preserves left multiplication by O. □

3 Recorded run and lattice

Definition 3.1 (Recorded particle trajectory data)**.** *The recorded data set is the tuple*

$$\mathscr{R} = \Big(N, \mathbb{T}, \{\mathcal{A}(t)\}_{t\in\mathbb{T}}, \{\mathcal{P}_t\}_{t\in\mathbb{T}}, \{x_i(t), v_i(t), \Phi_i(t)\}_{i\in\mathcal{I}, t\in\mathbb{T}}, \{K_\rho\}\Big).$$

Definition 3.2 (Site set and projections)**.** *For each pair $(i, t) \in \mathcal{I} \times \mathbb{T}$ define a lattice site $n_{i,t}$ and set*

$$\mathcal{N} := \{n_{i,t} \mid i \in \mathcal{I},\ t \in \mathbb{T}\}.$$

Define physical time, spatial, velocity, and spacetime projections by

$$\mathbf{t}(n_{i,t}) := t\, \Delta t,$$

$$\mathbf{x}(n_{i,t}) = x_i(t), \qquad \mathbf{v}(n_{i,t}) = v_i(t).$$
$$\Xi(n_{i,t}) := \big(\mathbf{t}(n_{i,t}), \mathbf{x}(n_{i,t})\big) \in \mathbb{R}^4.$$

All maps are well-defined by definition of $\mathcal{N}$.

Definition 3.3 (Oriented links)**.** *Define the interaction and temporal link sets by*

$$E_{\text{int}} := \{(n_{i,t} \to n_{j,t}) : (i,j) \in \mathcal{P}_t,\ t \in \mathbb{T}\},$$
$$E_{\text{time}} := \{(n_{i,t} \to n_{i,t+1}) : i \in \mathcal{I},\ 0 \le t < T\}.$$

For convenience, write

$$e_{\text{int}}(n_{i,t} \to n_{j,t}) := (n_{i,t} \to n_{j,t}), \qquad e_{\text{time}}(n_{i,t} \to n_{i,t+1}) := (n_{i,t} \to n_{i,t+1}).$$

By construction all source and target sites of E_{int} and E_{time} lie in $\mathcal{N}$ because $\mathcal{N}$ is defined for all $(i,t) \in \mathcal{I} \times \mathbb{T}$. Set

$$\mathcal{E} := E_{\text{int}} \cup E_{\text{time}}.$$

The lattice is $\mathcal{L} := (\mathcal{N}, \mathcal{E})$. For $e = (u \to v) \in \mathcal{E}$, write $s(e) := u$ and $t(e) := v$, define the spacetime displacement

$$\xi_e := \Xi(t(e)) - \Xi(s(e)) \in \mathbb{R}^4, \qquad \ell_e := \|\xi_e\|,$$

and define $e^{-1} := (t(e) \to s(e))$.

Proposition 3.4 (Finiteness and cardinality)**.** *All sets in Definitions 3.1 and 3.3 are finite. With $m_t := |\mathcal{P}_t|$,*

$$|E_{\text{int}}| = \sum_{t=0}^{T} m_t, \qquad |E_{\text{time}}| = NT, \qquad |\mathcal{E}| = NT + \sum_{t=0}^{T} m_t.$$

Proof. Finiteness follows from finiteness of $\mathbb{T}$ and all $\mathcal{P}_t$. The cardinality identity holds by one-to-one correspondence: each $(i,j,t) \in \bigcup_{t=0}^{T} \mathcal{P}_t \times \{t\}$ contributes one interaction link and each $(i,t) \in \mathcal{I} \times \{0, \dots, T-1\}$ contributes one temporal link. $\square$

4 Viscous force from interaction links

Definition 4.1 (Link degree and pairwise force coefficients)**.** *For each (i,t) define the raw link degree and normalized weights*

$$\deg_{\text{link}}(i,t) := \sum_{(i,j)\in\mathcal{P}_t} K_\rho(x_i(t), x_j(t)),$$

$$\omega_{ij}^{\text{link}}(t) := \begin{cases} \dfrac{K_\rho(x_i(t), x_j(t))}{\deg_{\text{link}}(i,t)}, & (i,j) \in \mathcal{P}_t,\ \deg_{\text{link}}(i,t) > 0, \\ 0, & \textit{otherwise.} \end{cases}$$

Definition 4.2 (Viscous pair force and site force)**.** *For each interaction link $e = (n_{i,t} \to n_{j,t}) \in E_{\text{int}}$ define*

$$\mathbf{F}^{\text{visc}}(e) := \nu_{\text{visc}} \omega_{ij}^{\text{link}}(t)\,(v_j(t) - v_i(t)) \in \mathbb{R}^3.$$

For $x = n_{i,t} \in \mathcal{N}$ define

$$\mathbf{F}^{\text{visc}}(x) := \sum_{\substack{e \in E_{\text{int}} \\ s(e) = x}} \mathbf{F}^{\text{visc}}(e).$$

If the sum is over an empty set, it is 0.

Lemma 4.3 (Weight properties). *For every $i \in \mathcal{A}(t)$ and $t \in \mathbb{T}$,*

$$\omega_{ij}^{\text{link}}(t) \geq 0, \qquad \sum_{j:(i,j)\in\mathcal{P}_t} \omega_{ij}^{\text{link}}(t) = \mathbf{1}_{\{\deg_{\text{link}}(i,t)>0\}}.$$

Proof. Positivity is immediate from $K_\rho \geq \kappa_* > 0$. If $\deg_{\text{link}}(i,t) > 0$, summing the definitions over j gives

$$\sum_{j:(i,j)\in\mathcal{P}_t} \omega_{ij}^{\text{link}}(t) = \frac{1}{\deg_{\text{link}}(i,t)} \sum_{j:(i,j)\in\mathcal{P}_t} K_\rho(x_i(t), x_j(t)) = 1.$$

If $\deg_{\text{link}}(i,t) = 0$, every summand is 0 by convention. □

Lemma 4.4 (Node reconstruction of viscous force). *For every $x = n_{i,t} \in \mathcal{N}$,*

$$\mathbf{F}^{\text{visc}}(x) = \sum_{\substack{e\in E_{\text{int}} \\ s(e)=x}} \mathbf{F}^{\text{visc}}(e).$$

Proof. By Definition 3.3, the interaction links with source $n_{i,t}$ in E_{int} are exactly $\{(n_{i,t} \to n_{j,t}) : (i,j) \in \mathcal{P}_t\}$, which is the same index set used in Definition 4.2. □

Proposition 4.5 (Orthogonal basis covariance). *Let $O \in O(3)$ and consider transformed data $x_i'(t) = Ox_i(t)$, $v_i'(t) = Ov_i(t)$. For every site $x \in \mathcal{N}$,*

$$\mathbf{F}^{\text{visc}}(x; \textit{transformed data}) = O\,\mathbf{F}^{\text{visc}}(x; \textit{original data}).$$

Moreover, real orthogonal matrices satisfy $O^\top = O^{-1}$, hence $O^\dagger = O^{-1}$, so $O \in U(3)$. Consequently, orthogonal changes of the physical coordinate frame embed in unitary changes of the complexified color index.

Proof. Covariance of the force is Lemma 2.3. For the embedding: since O has real entries, $O^\dagger = O^\top = O^{-1}$, so $O^\dagger O = I$ and O is unitary when viewed as a complex matrix. □

5 Color triplets and $SU(3)$ link transport

Definition 5.1 (Normalized color vectors). *Fix constants $g_3 > 0$, $\ell_0 > 0$, $\hbar_{\text{eff}} > 0$, $\mu_p > 0$, $\epsilon_c > 0$. For $x = n_{i,t} \in \mathcal{N}$ and each coordinate $\alpha \in \{1,2,3\}$, define*

$$\tilde{c}_x^{(\alpha)} := (\mathbf{F}^{\text{visc}}(x))_\alpha \exp\left(i\frac{\mu_p\, v_i^{(\alpha)}(t)\ell_0}{\hbar_{\text{eff}}}\right),$$

$$\mathbf{c}_x := \frac{\tilde{\mathbf{c}}_x}{\sqrt{\tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x + \epsilon_c^2}} \in \mathbb{C}^3.$$

Theorem 5.2 (Emergent local $SU(3)$ color symmetry). *For any map $\Omega : \mathcal{N} \to O(3)$ and each site $x \in \mathcal{N}$, define*

$$\mathbf{c}_x^\Omega := \Omega(x)\mathbf{c}_x.$$

Because $\Omega(x)$ is real orthogonal, $\|\mathbf{c}_x^\Omega\| = \|\mathbf{c}_x\|$ for every $x \in \mathcal{N}$. Hence the color data carry a local $O(3)$ redundancy, which embeds as real unitaries into $U(3)$. Imposing $\det\Omega(x) = 1$ for every site yields a local $SO(3)$ subgroup $SO(3) \subset SU(3)$ on each site. The full local $SU(3)$ bundle symmetry used below is then defined explicitly by $\Omega : \mathcal{N} \to SU(3)$ in Definition 6.2.

Proof.

$$\mathbf{c}_x^{\Omega} = \frac{\Omega(x)\tilde{\mathbf{c}}_x}{\sqrt{\tilde{\mathbf{c}}_x^{\dagger}\tilde{\mathbf{c}}_x + \epsilon_c^2}} = \Omega(x)\mathbf{c}_x.$$

Since $\Omega(x)$ is orthogonal,

$$\|\mathbf{c}_x^{\Omega}\|^2 = \frac{\tilde{\mathbf{c}}_x^{\dagger}\Omega(x)^{\dagger}\Omega(x)\tilde{\mathbf{c}}_x}{\tilde{\mathbf{c}}_x^{\dagger}\tilde{\mathbf{c}}_x + \epsilon_c^2} = \frac{\tilde{\mathbf{c}}_x^{\dagger}\tilde{\mathbf{c}}_x + \epsilon_c^2}{\tilde{\mathbf{c}}_x^{\dagger}\tilde{\mathbf{c}}_x + \epsilon_c^2} = \|\mathbf{c}_x\|^2.$$

Therefore the normalization denominator is unchanged and $\|\mathbf{c}_x^{\Omega}\| = \|\mathbf{c}_x\|$. □

Remark 5.3 (Physical interpretation of the color construction). The factor $\exp\big(i\mu_p v_i^{(\alpha)}\ell_0/\hbar_{\text{eff}}\big)$ is a discrete analogue of the de Broglie phase $e^{ip\ell/\hbar}$: it encodes the kinematic momentum of particle i in the α-direction into the phase of the α-th color component. Nodes moving in different directions therefore carry genuinely different internal phases, even if they experience the same force magnitude. This momentum-phase coupling ensures that the resulting parallel transport $U_3(e)$ depends not only on the local force profile but also on how the color triplet changes from one site to the next along the oriented edge.

The regularization $\epsilon_c > 0$ ensures that $\mathbf{c}_x$ is well-defined even when the viscous force vanishes, e.g. for an isolated particle or when $\nu_{\text{visc}} = 0$. If $\mathbf{c}_x = 0$, then

$$\Re\big(\mathbf{c}_x^{\dagger}\lambda_a(\mathbf{c}_y - \mathbf{c}_x)\big) = 0$$

for all outgoing links from x, hence $U_3(x \to y) = I$ for all such links. Thus nodes with trivial source color couple trivially through this link transport. When $\mathbf{c}_y = 0$ and $\mathbf{c}_x \neq 0$, the link need not be trivial: it records the local decay of the color field along the edge.

The normalization satisfies $\|\mathbf{c}_x\|^2 < 1$ for all x (Lemma 5.4). Combined with $|\lambda_a|_{\text{op}} \leq 2$ for all Gell-Mann generators, this gives the operator bound

$$\|\Theta_e\|_{\text{op}} \leq \frac{g_3}{2}\sum_{a=1}^{8}\big|\Re\big(\mathbf{c}_x^{\dagger}\lambda_a(\mathbf{c}_y - \mathbf{c}_x)\big)\big|\|\lambda_a\|_{\text{op}} \leq \frac{g_3}{2}\cdot 8\cdot\|\mathbf{c}_x\|\,\|\mathbf{c}_y - \mathbf{c}_x\|\cdot 2 < 16g_3,$$

so the gauge coupling strength is controlled uniformly over all links and runs.

Lemma 5.4 (Normalization of color vectors)**.** *For every site $x \in \mathcal{N}$,*

$$\mathbf{c}_x^{\dagger}\mathbf{c}_x = \frac{\tilde{\mathbf{c}}_x^{\dagger}\tilde{\mathbf{c}}_x}{\tilde{\mathbf{c}}_x^{\dagger}\tilde{\mathbf{c}}_x + \epsilon_c^2} \in [0, 1),$$

so $\mathbf{c}_x$ is well-defined and $\|\mathbf{c}_x\| < 1$.

Proof. Since $\epsilon_c > 0$, the denominator $\tilde{\mathbf{c}}_x^{\dagger}\tilde{\mathbf{c}}_x + \epsilon_c^2$ is strictly positive. The ratio $r/(r + \epsilon_c^2)$ for $r = \tilde{\mathbf{c}}_x^{\dagger}\tilde{\mathbf{c}}_x \geq 0$ lies in $[0, 1)$, with the upper bound strict because $\epsilon_c^2 > 0$. □

Lemma 5.5 (Color reparametrization)**.** *Let $\Omega : \mathcal{N} \to SU(3)$ be an arbitrary map. Define $\mathbf{c}_x^{\Omega} := \Omega(x)\mathbf{c}_x$ for each $x \in \mathcal{N}$. Then $(\mathbf{c}_x^{\Omega})^{\dagger}\mathbf{c}_x^{\Omega} = \mathbf{c}_x^{\dagger}\mathbf{c}_x$ and $\|\mathbf{c}_x^{\Omega}\| = \|\mathbf{c}_x\|$.*

Proof. Since $\Omega(x) \in U(3)$, $\Omega(x)^{\dagger}\Omega(x) = I$. Hence

$$(\mathbf{c}_x^{\Omega})^{\dagger}\mathbf{c}_x^{\Omega} = \mathbf{c}_x^{\dagger}\Omega(x)^{\dagger}\Omega(x)\mathbf{c}_x = \mathbf{c}_x^{\dagger}\mathbf{c}_x.$$

□

Definition 5.6 (Gell-Mann generators). *The eight Gell-Mann matrices $\lambda_a \in \mathbb{C}^{3\times 3}$, $a = 1, \dots, 8$, are*

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Each λ_a is Hermitian and $\operatorname{tr} \lambda_a = 0$. The generators $\{i\lambda_a/2\}_{a=1}^{8}$ form a basis of the Lie algebra $\mathfrak{su}(3)$ of traceless skew-Hermitian matrices. We write

$$\mathfrak{h}_3 := \{H \in \mathbb{C}^{3\times 3} : H^\dagger = H,\ \operatorname{tr} H = 0\}$$

for the corresponding Hermitian generator space, so that $i\mathfrak{h}_3 = \mathfrak{su}(3)$.

Lemma 5.7 (Gell-Mann orthogonality). *The Gell-Mann matrices satisfy*

$$\operatorname{tr}(\lambda_a \lambda_b) = 2\delta_{ab}, \qquad a, b \in \{1, \dots, 8\}.$$

Proof. Direct matrix multiplication: for $a = b$ each diagonal entry of λ_a^2 contributes, giving trace 2; for $a \neq b$ the off-diagonal structure ensures the trace vanishes. This is a finite componentwise check whose result is standard. □

Lemma 5.8 (Hermitian traceless span). *If $\alpha_1, \dots, \alpha_8 \in \mathbb{R}$, then $H := \sum_{a=1}^{8} \alpha_a \lambda_a$ is Hermitian and traceless.*

Proof. Each λ_a is Hermitian and traceless by Definition 5.6. Real linear combinations preserve both properties. □

Definition 5.9 (Generator and transport on links). *For each oriented link $e = (x \to y) \in \mathcal{E}$ define the algebra element*

$$\mathcal{A}_e := \frac{1}{2} \sum_{a=1}^{8} \Re\big(\mathbf{c}_x^\dagger \lambda_a (\mathbf{c}_y - \mathbf{c}_x)\big) \lambda_a \in \mathfrak{h}_3, \qquad \Theta_e := g_3 \, \mathcal{A}_e.$$

and the $SU(3)$ link transport

$$U_3(e) := \exp(i\Theta_e) \in SU(3), \qquad U_3(e^{-1}) := U_3(e)^{-1}.$$

Remark 5.10 (Justification of the transport construction). The formula

$$\mathcal{A}_e = \frac{1}{2} \sum_a \Re\big(\mathbf{c}_x^\dagger \lambda_a (\mathbf{c}_y - \mathbf{c}_x)\big) \lambda_a$$

is the orthogonal projection of the discrete color increment $\mathbf{c}_y - \mathbf{c}_x$ against the source color vector onto the Hermitian traceless subspace spanned by $\{\lambda_a\}_{a=1}^{8}$. To see this, recall that the Hermitian traceless matrices carry the inner product $\langle A, B\rangle = \frac{1}{2}\operatorname{tr}(AB)$, under which $\{\lambda_a\}$ is an orthogonal basis with $\langle \lambda_a, \lambda_b\rangle = \delta_{ab}$ by Lemma 5.7. The a-th component of the projected discrete derivative is

$$\left\langle \lambda_a, \tfrac{1}{2}\big(\mathbf{c}_x(\mathbf{c}_y - \mathbf{c}_x)^\dagger + (\mathbf{c}_y - \mathbf{c}_x)\mathbf{c}_x^\dagger\big)\right\rangle = \frac{1}{2}\operatorname{tr}\Big(\lambda_a\big(\mathbf{c}_x(\mathbf{c}_y - \mathbf{c}_x)^\dagger + (\mathbf{c}_y - \mathbf{c}_x)\mathbf{c}_x^\dagger\big)\Big) = \Re\big(\mathbf{c}_x^\dagger \lambda_a(\mathbf{c}_y - \mathbf{c}_x)\big),$$

where we used $\lambda_a^\dagger = \lambda_a$. Since the resulting coefficients are real, $\mathcal{A}_e$ is exactly this projection. The coupling is $\Theta_e = g_3\mathcal{A}_e$.

The factor $g_3/2$ is chosen to match the continuum covariant derivative $D_\mu = \partial_\mu - ig_3 A_\mu^a \lambda_a/2$: if the color field varies smoothly and the edge displacement is small, then $\mathbf{c}_y - \mathbf{c}_x$ is first order in the edge length, so the transport linearizes to

$$U_3(e) = I + i\Theta_e + O(g_3^2) = I + ig_3\,\mathcal{A}_e + O(g_3^2) = I + \frac{ig_3}{2}\sum_a \Re\big(\mathbf{c}_x^\dagger\lambda_a(\mathbf{c}_y - \mathbf{c}_x)\big)\lambda_a + O(g_3^2),$$

which is the first-order approximation to a continuum parallel transporter $P\exp(ig_3\int A_\mu\, dx^\mu)$.

Proposition 5.11 (Link transport is $SU(3)$)**.** *For every $e \in \mathcal{E}$, $U_3(e) \in SU(3)$ and $U_3(e^{-1}) = U_3(e)^{-1}$.*

Proof. Each coefficient $\Re(\mathbf{c}_x^\dagger\lambda_a(\mathbf{c}_y - \mathbf{c}_x))$ is real. By Lemma 5.8, Θ_e is Hermitian and traceless, so $i\Theta_e$ is skew-Hermitian and traceless. Hence $U_3(e) = \exp(i\Theta_e)$ is unitary. Since $\operatorname{tr}\Theta_e = 0$, the identity $\det\exp(M) = \exp(\operatorname{tr} M)$ gives $\det U_3(e) = \exp(i\operatorname{tr}\Theta_e) = 1$. Therefore $U_3(e) \in SU(3)$. The inverse relation $U_3(e^{-1}) = U_3(e)^{-1}$ is imposed by convention. □

Remark 5.12. There are no oriented self-loops in $\mathcal{E}$. Interaction links exclude (i, i) by definition, and temporal links connect distinct times t and $t+1$. If one formally assigns the link transport formula to a self-loop, it still lands in $SU(3)$ because $\mathcal{A}_e$ is Hermitian and traceless.

6 Holonomy and gauge invariance

Definition 6.1 (Path holonomy)**.** *For an oriented path $\gamma = (v_0 \to v_1, \ldots, v_{m-1} \to v_m)$ with $v_k \in \mathcal{N}$, define the ordered product (left-to-right composition)*

$$U_3(\gamma) := U_3(v_0 \to v_1)\, U_3(v_1 \to v_2) \cdots U_3(v_{m-1} \to v_m) = \prod_{k=1}^{m} U_3(v_{k-1} \to v_k).$$

This ordering convention uses the usual link ordering in lattice gauge theory, where $U_\mu(x)$ transports from site x to site $x+\hat{\mu}$. If $v_m = v_0$, γ is a loop and we define the Wilson loop trace $W_3(\gamma) := \operatorname{tr} U_3(\gamma)$.

Definition 6.2 (Gauge map)**.** *A gauge map is any map $\Omega : \mathcal{N} \to SU(3)$. For a link $e = (x \to y) \in \mathcal{E}$ define the gauge-transformed transport*

$$U_3^\Omega(e) := \Omega(y)\, U_3(e)\, \Omega(x)^{-1}.$$

Transformed color vectors are $\mathbf{c}_x^\Omega := \Omega(x)\mathbf{c}_x$.

Proposition 6.3 (Gauge invariance of Wilson traces)**.** *For any closed loop $\gamma = (v_0, \ldots, v_m = v_0)$ and any gauge map Ω,*

$$\operatorname{tr}\left(U_3^\Omega(\gamma)\right) = \operatorname{tr}\left(U_3(\gamma)\right).$$

Proof. Compute the gauge-transformed holonomy:

$$U_3^{\Omega}(\gamma) = \prod_{k=1}^{m} \Omega(v_k)\, U_3(v_{k-1} \to v_k)\, \Omega(v_{k-1})^{-1}.$$

Consecutive factors telescope: $\Omega(v_k)$ from step k cancels $\Omega(v_k)^{-1}$ from step $k+1$, since v_k appears as the target of step k and the source of step $k+1$. After all m steps, the remaining factors are $\Omega(v_m) = \Omega(v_0)$ on the left and $\Omega(v_0)^{-1}$ on the right, giving

$$U_3^{\Omega}(\gamma) = \Omega(v_0)\, U_3(\gamma)\, \Omega(v_0)^{-1}.$$

Trace cyclicity then gives $\mathrm{tr}(U_3^{\Omega}(\gamma)) = \mathrm{tr}(\Omega(v_0)^{-1}\Omega(v_0)\, U_3(\gamma)) = \mathrm{tr}(U_3(\gamma))$. $\square$

Definition 6.4 (Canonical elementary plaquettes)**.** *For each time slice $0 \le t < T$ and each ordered pair $(i,j) \in \mathcal{P}_t \cap \mathcal{P}_{t+1}$, define the elementary plaquette*

$$P_{ij,t} := (n_{i,t} \to n_{i,t+1} \to n_{j,t+1} \to n_{j,t} \to n_{i,t}).$$

Its boundary consists of one forward temporal link for particle i, one forward interaction link at time $t+1$, one backward temporal link for particle j, and one backward interaction link at time t. Set

$$\mathcal{C}_{\square} := \{P_{ij,t} :\ (i,j) \in \mathcal{P}_t \cap \mathcal{P}_{t+1},\ 0 \le t < T\}.$$

Definition 6.5 (Plaquette geometry)**.** *For $P = P_{ij,t} \in \mathcal{C}_{\square}$, define the base point*

$$z_P := \Xi(n_{i,t}) \in \mathbb{R}^4,$$

the temporal and interaction edge displacements at the base point

$$\tau_{i,t} := \Xi(n_{i,t+1}) - \Xi(n_{i,t}) \in \mathbb{R}^4, \qquad \sigma_{ij,t} := \Xi(n_{j,t}) - \Xi(n_{i,t}) \in \mathbb{R}^4,$$

and the oriented area bivector

$$\Sigma_P^{\mu\nu} := \tau_{i,t}^{\mu}\sigma_{ij,t}^{\nu} - \tau_{i,t}^{\nu}\sigma_{ij,t}^{\mu}.$$

Its area is

$$A_P := |\Sigma_P| := \left(\frac{1}{2} \sum_{\mu,\nu=0}^{3} \Sigma_P^{\mu\nu}\Sigma_P^{\mu\nu} \right)^{1/2}.$$

Definition 6.6 (Plaquette holonomy and discrete field strength)**.** *For $P = P_{ij,t} \in \mathcal{C}_{\square}$, define the plaquette holonomy by the ordered product*

$$U_3(P) := U_3(n_{i,t} \to n_{i,t+1})\, U_3(n_{i,t+1} \to n_{j,t+1})\, U_3(n_{j,t+1} \to n_{j,t})\, U_3(n_{j,t} \to n_{i,t}).$$

Whenever $A_P > 0$ and $U_3(P)$ lies in the principal logarithm domain, define the discrete plaquette field strength by

$$F_P := \frac{1}{i g_3 A_P} \log U_3(P) \in \mathfrak{h}_3.$$

Definition 6.7 (Wilson action)**.** *Set $\beta := 6/g_3^2$. The Wilson plaquette action is*

$$S_{\mathrm{W}} := \beta \sum_{P \in \mathcal{C}_{\square}} \left(1 - \frac{1}{3} \Re\, \mathrm{tr}\, U_3(P) \right).$$

Remark 6.8. The normalization $1 - \frac{1}{3}\Re\operatorname{tr} U_3(P)$ vanishes when $U_3(P) = I$ and is non-negative for all $U_3(P) \in SU(3)$, since $|\operatorname{tr} U|/3 \le 1$ by the triangle inequality applied to the three eigenvalues of U. In the $N \to \infty$ thermodynamic limit, Theorem 9.13 shows that the sum of these plaquette terms converges to the mean-field Wilson action S_{W}^{∞} defined on the stationary measure π_∞.

Proposition 6.9 (Gauge invariance of Wilson action)**.** *For any gauge map* Ω*,* $S_{\mathrm{W}}^{\Omega} = S_{\mathrm{W}}$*.*

Proof. Each plaquette trace $\Re\operatorname{tr} U_3(P)$ is invariant under gauge transformation by Proposition 6.3. The sum and the constant term are unchanged, so $S_{\mathrm{W}}^{\Omega} = S_{\mathrm{W}}$. □

7 Matter sector on the same lattice

Definition 7.1 (Matter fields and actions)**.** *For each site* $x \in \mathcal{N}$ *choose a fermionic spinor* $\psi_x \in \mathbb{C}^3$*, its conjugate* $\bar\psi_x \in (\mathbb{C}^3)^\dagger$*, and a real scalar* $\phi_x \in \mathbb{R}$*. Fix* $m_F > 0$*, and let* $m_\phi > 0$ *be the scalar witness mass from Definition 9.17. For* $e = (x \to y) \in \mathcal{E}$ *set*

$$\ell_{xy} := \|\Xi(y) - \Xi(x)\| + \eta = \ell_e + \eta$$

with $\eta > 0$*. The covariant fermionic action is*

$$S_{\mathrm{F}} := \sum_{e\in\mathcal{E}} \bar\psi_{t(e)}\left(\psi_{t(e)} - U_3(e)\psi_{s(e)}\right) + m_F \sum_{x\in\mathcal{N}} \bar\psi_x\psi_x,$$

and the covariant scalar action is

$$S_\phi := \frac{1}{2}\sum_{e\in\mathcal{E}} \frac{(\phi_{t(e)} - \phi_{s(e)})^2}{\ell_{xy}} + \frac{m_\phi^2}{2}\sum_{x\in\mathcal{N}} \phi_x^2.$$

The gauge group acts on matter fields by

$$\psi_x \mapsto \Omega(x)\psi_x, \qquad \bar\psi_x \mapsto \bar\psi_x\Omega(x)^{-1}, \qquad \phi_x \mapsto \phi_x.$$

Remark 7.2 (Distance weighting in the scalar action)*.* The factor $1/\ell_{xy}$ in the scalar kinetic term is a discrete analogue of the continuum normalization $(\nabla_\mu\phi)^2$. In the continuum, the square of a derivative has dimensions $[\phi]^2/[\text{length}]^2$, so a finite difference $(\phi_y - \phi_x)^2$ must be divided by $\|\mathbf{x}(y) - \mathbf{x}(x)\|$ (one power of length rather than two) for a first-order finite-difference scheme. In the spacetime lattice used here, the relevant finite-difference length is the four-dimensional edge norm $\|\Xi(y) - \Xi(x)\|$. Specifically, if $\phi_{t(e)} - \phi_{s(e)} \approx (\partial_\mu\phi)\,\xi_e^\mu$ with $\|\Xi(y) - \Xi(x)\| \approx \ell_{xy}$, then $(\phi_{t(e)} - \phi_{s(e)})^2/\ell_{xy} \approx \ell_{xy}\,|\nabla\phi|^2$, and summing over a regular lattice with spacing $a = \ell_{xy}$ recovers $a^d\,|\nabla\phi|^2$ as in the usual scalar lattice action.

The parameter $\eta > 0$ prevents division by zero for coincident site positions (self-loops, or particles occupying the same spacetime point). In practice $\eta \ll \ell_{\mathrm{visc}}$ so that $\ell_{xy} \approx \|\Xi(y) - \Xi(x)\|$ for all non-coincident links.

Proposition 7.3 (Matter gauge invariance)**.** *Under the gauge action of Definition 7.1,*

$$S_{\mathrm{F}}^{\Omega} = S_{\mathrm{F}}, \qquad S_\phi^{\Omega} = S_\phi.$$

Proof. For each fermionic link term,

$$\bar\psi_{t(e)}\big(\psi_{t(e)} - U_3(e)\psi_{s(e)}\big) \mapsto \bar\psi_{t(e)}\Omega(t(e))^{-1}\big(\Omega(t(e))\psi_{t(e)} - \Omega(t(e))U_3(e)\Omega(s(e))^{-1}\Omega(s(e))\psi_{s(e)}\big)$$

$$= \bar{\psi}_{t(e)}\big(\psi_{t(e)} - U_3(e)\psi_{s(e)}\big)\,,$$

where we used $\Omega(t(e))^{-1}\Omega(t(e)) = I$ and $\Omega(s(e))^{-1}\Omega(s(e)) = I$. The mass term $m_F\bar{\psi}_x\psi_x$ is invariant because $\Omega(x)^{-1}\Omega(x) = I$ pointwise. The scalar action S_ϕ depends only on scalar fields, which are gauge-invariant, so $S_\phi^\Omega = S_\phi$. $\square$

Remark 7.4 (Matter sector on the lattice). The same finite link set $\mathcal{E}$ supports both matter sectors. The fermionic term couples oriented transport through $U_3(e)$, while the scalar term uses the same link differences as a discrete kinetic energy. The scalar sector is the free massive witness field canonically tied below to the stationary mean-field stiffness scale of the companion convergence paper.

8 Finite lattice-QFT data package

Theorem 8.1 (Finite $SU(3)$ data package)**.** *Given a run $\mathscr{R}$ as in Definition 3.1 and the constructions in Sections 3–7, the tuple*

$$\mathcal{Q} = \Big(T, \Delta t, \{\mathcal{A}(t)\}, \{\mathcal{P}_t\}, \mathcal{N}, E_{\text{int}}, E_{\text{time}}, \mathcal{E}, \mathcal{C}_\square, \{x_i(t), v_i(t), \Phi_i(t)\}_{i,t}, K_\rho, \nu_{\text{visc}}, \hbar_{\text{eff}}, \mu_p, \ell_0, \epsilon_c, g_3, \beta, \eta, \{\Xi(x)\}_{x\in\mathcal{N}}, \{\Sigma$$

is finite and explicit. It carries:

1. *a finite spacetime lattice $(\mathcal{N}, \mathcal{E})$ with interaction and temporal links;*
2. *a finite canonical plaquette family $\mathcal{C}_\square$ for the Wilson action;*
3. *explicit plaquette geometry (z_P, Σ_P, A_P) and plaquette holonomies $U_3(P)$;*
4. *$SU(3)$ parallel transport maps $\{U_3(e)\}_{e\in\mathcal{E}}$;*
5. *gauge-invariant Wilson plaquette traces and Wilson action S_{W};*
6. *gauge-covariant fermionic and scalar matter actions S_{F}, S_ϕ.*

All objects required for a subsequent axiom-verification paper are present.

Proof. Finiteness of $(\mathcal{N}, \mathcal{E})$ is Proposition 3.4. Finiteness of $\mathcal{C}_\square$ follows from Definition 6.4 because it is indexed by the finite set of triples (i, j, t) with $0 \le t < T$ and $(i, j) \in \mathcal{P}_t \cap \mathcal{P}_{t+1}$. The spacetime embedding, area bivectors, and plaquette holonomies are finite families because they are explicit algebraic functions of $\mathcal{N}$ and $\mathcal{C}_\square$. Finiteness of $\{U_3(e)\}$ follows because $\mathcal{E}$ is finite and each $U_3(e)$ is defined by a finite matrix exponential. The actions S_{W}, S_{F}, S_ϕ are finite sums over finite index sets by the same finiteness. Gauge invariance of S_{W} is Proposition 6.9; gauge invariance of S_{F} and S_ϕ is Proposition 7.3. $\square$

9 Discussion

9.1 Lattice-gauge interpretation

The lattice $(\mathcal{N}, \mathcal{E})$ is finite and data dependent. The site set is a finite subset of spacetime determined by the particle data, while the link set contains both temporal worldline links and same-time interaction links. The link variables $U_3(e)$ are not independent: they are *derived* from the trajectory data via the color construction of Definitions 5.1–5.9. The gauge group, the coupling strength g_3, and the parallel transport all emerge from the viscous dynamics rather than being postulated.

The Wilson action is written on the canonical plaquette family $\mathcal{C}_\Box$ from Definition 6.4, and the matter actions use the same spacetime links. All gauge-sector observables are therefore evaluated on explicit elementary 2-cells rather than on an arbitrary loop family.

9.2 Thermodynamic limit of graph observables

The convergence paper proves exchangeability of the invariant laws, a finite de Finetti representation for the empirical measure, collapse of the mixing measure to the unique stationary mean-field law π_∞, and the same N-uniform logarithmic Sobolev constant in the limit [1]. These results are exactly the input needed to pass the discrete interaction graph itself to the stationary mean-field regime.

Definition 9.1 (Pairwise link rule and empirical edge measure)**.** *Fix a recorded time slice $t \in \mathbb{T}$ and, for the N-particle system, write*

$$Z_i^N(t) := (X_i^N(t), V_i^N(t)) \in \mathbb{R}^6, \qquad L_t^N := \frac{1}{N}\sum_{i=1}^N \delta_{Z_i^N(t)}.$$

Assume that the same-time link set $\mathcal{P}_t^N$ is generated by the same label-invariant pair rule for every N, in the sense that there exists a bounded measurable indicator

$$\chi_t : \mathbb{R}^6 \times \mathbb{R}^6 \to \{0, 1\}$$

such that

$$\mathbf{1}_{\{(i,j)\in\mathcal{P}_t^N\}} = \chi_t(Z_i^N(t), Z_j^N(t)) \qquad (i \neq j).$$

Define the empirical edge measure by

$$\Gamma_t^N := \frac{1}{N^2}\sum_{i\neq j} \mathbf{1}_{\{(i,j)\in\mathcal{P}_t^N\}}\delta_{(Z_i^N(t), Z_j^N(t))}.$$

Theorem 9.2 (Stationary thermodynamic limit of the interaction graph)**.** *Assume the stationary thermodynamic limit from the companion convergence paper: under the invariant law of the N-particle system, the random empirical measure L_t^N converges in law to the deterministic stationary mean-field law π_∞ as $N \to \infty$. Let $\Phi : \mathbb{R}^6 \times \mathbb{R}^6 \to \mathbb{R}$ be bounded and continuous, and assume that $\chi_t\Phi$ is continuous at $(\pi_\infty \otimes \pi_\infty)$-almost every pair. Then*

$$\int_{\mathbb{R}^6\times\mathbb{R}^6} \Phi \, d\Gamma_t^N \xrightarrow[N\to\infty]{\mathbb{P}} \iint_{\mathbb{R}^6\times\mathbb{R}^6} \chi_t(z, z')\, \Phi(z, z')\, \pi_\infty(dz)\, \pi_\infty(dz').$$

Equivalently,

$$\frac{1}{N^2}\sum_{(i,j)\in\mathcal{P}_t^N} \Phi(Z_i^N(t), Z_j^N(t)) \xrightarrow[N\to\infty]{\mathbb{P}} \iint_{\mathbb{R}^6\times\mathbb{R}^6} \chi_t(z, z')\, \Phi(z, z')\, \pi_\infty(dz)\, \pi_\infty(dz').$$

If the number of recorded slices $T + 1$ is fixed, then the same convergence holds after summing over $t \in \mathbb{T}$.

Proof. Write

$$\Psi_t(z,z') := \chi_t(z,z')\,\Phi(z,z').$$

By Definition 9.1,

$$\begin{aligned}\int \Phi\, d\Gamma_t^N &= \frac{1}{N^2}\sum_{i\neq j}\Psi_t(Z_i^N(t),Z_j^N(t))\\ &= \frac{1}{N^2}\sum_{i,j}\Psi_t(Z_i^N(t),Z_j^N(t)) - \frac{1}{N^2}\sum_{i=1}^{N}\Psi_t(Z_i^N(t),Z_i^N(t))\\ &= \iint_{\mathbb{R}^6\times\mathbb{R}^6}\Psi_t(z,z')\,L_t^N(dz)\,L_t^N(dz') - R_t^N,\end{aligned}$$

where

$$|R_t^N| \le \frac{\|\Psi_t\|_\infty}{N}.$$

Hence $R_t^N \to 0$ in probability.

Because L_t^N converges in law to the deterministic measure π_∞, the product measure $L_t^N \otimes L_t^N$ converges in law to $\pi_\infty \otimes \pi_\infty$ by the continuous mapping theorem. The boundedness and almost-everywhere continuity assumption on Ψ_t then give

$$\iint \Psi_t(z,z')\,L_t^N(dz)\,L_t^N(dz') \xrightarrow[N\to\infty]{\mathbb{P}} \iint \Psi_t(z,z')\,\pi_\infty(dz)\,\pi_\infty(dz').$$

Combining the last two displays proves the first claim. The displayed sum over $(i,j)\in\mathcal{P}_t^N$ is exactly the same quantity written in index notation, so the equivalent form follows. If $T+1$ is fixed, summing the convergences over finitely many times preserves convergence in probability. □

Corollary 9.3 (Thermodynamic limit of weighted link sums)**.** *Let*

$$W_t : \mathbb{R}^6\times\mathbb{R}^6\times\mathcal{P}(\mathbb{R}^6)\to\mathbb{R}$$

be bounded, and assume that W_t is jointly continuous at every point (z,z',π_∞) for which $\chi_t(z,z')=1$. Then for every bounded continuous Φ,

$$\frac{1}{N^2}\sum_{(i,j)\in\mathcal{P}_t^N} W_t(Z_i^N(t),Z_j^N(t);L_t^N)\,\Phi(Z_i^N(t),Z_j^N(t)) \xrightarrow[N\to\infty]{\mathbb{P}} \iint \chi_t(z,z')\,W_t(z,z';\pi_\infty)\,\Phi(z,z')\,\pi_\infty(dz)\,\pi_\infty(dz').$$

Proof. Set

$$\Psi_t^N(z,z') := \chi_t(z,z')\,W_t(z,z';L_t^N)\,\Phi(z,z'), \qquad \Psi_t^\infty(z,z') := \chi_t(z,z')\,W_t(z,z';\pi_\infty)\,\Phi(z,z').$$

Since $L_t^N \to \pi_\infty$ in law and the limit is deterministic, $W_t(z,z';L_t^N)\to W_t(z,z';\pi_\infty)$ in probability for each continuity point (z,z',π_∞). Boundedness then lets us pass to the limit in the same decomposition used in Theorem 9.2, with Ψ_t^N in place of Ψ_t. The diagonal contribution is still $O(N^{-1})$, so the result follows. □

Remark 9.4. Theorem 9.2 is the thermodynamic step needed to replace discrete interaction-link sums by mean-field integrals without introducing any external discretization of the limiting law. To pass the Wilson action itself to the thermodynamic limit one also needs a plaquette-level statement, because the canonical plaquettes depend on two adjacent time slices.

Definition 9.5 (Adjacent-time empirical path and plaquette measures)**.** *Fix $0 \le t < T$ and write*

$$\widetilde{Z}_i^N(t) := \big(Z_i^N(t), Z_i^N(t+1)\big) \in \mathbb{R}^{12}, \qquad M_t^N := \frac{1}{N}\sum_{i=1}^N \delta_{\widetilde{Z}_i^N(t)}.$$

Define the persistent-plaquette indicator by

$$\chi_{\square,t}\big((z_0,z_1),(w_0,w_1)\big) := \chi_t(z_0,w_0)\,\chi_{t+1}(z_1,w_1),$$

and the empirical plaquette measure by

$$\Gamma_{\square,t}^N := \frac{1}{N^2}\sum_{i\neq j} \chi_{\square,t}\big(\widetilde{Z}_i^N(t), \widetilde{Z}_j^N(t)\big)\,\delta_{\big(\widetilde{Z}_i^N(t),\widetilde{Z}_j^N(t)\big)}.$$

Definition 9.6 (Stationary mean-field path law)**.** *For every finite horizon $S > 0$, let $\Pi_{[0,S]}^\infty$ be the law on $C([0,S];\mathbb{R}^6)$ of the unique McKean–Vlasov solution with initial law π_∞. Because π_∞ is stationary, $\Pi_{[0,S]}^\infty$ is stationary under time shifts. For each integer $0 \le t < S$, define the adjacent-time mean-field law by*

$$\Lambda_t^\infty := (\mathrm{ev}_{t,t+1})_\# \Pi_{[0,S]}^\infty, \qquad \mathrm{ev}_{t,t+1}(\gamma) := \big(\gamma(t), \gamma(t+1)\big).$$

Theorem 9.7 (Stationary path-space propagation of chaos)**.** *For every finite horizon $S > 0$, let $\widehat{P}_{[0,S]}^N$ be the law on $C([0,S];(\mathbb{R}^6)^N)$ of the N-particle kinetic system started from its invariant law π_N, and define the empirical path measure*

$$\mathfrak{M}_{[0,S]}^N := \frac{1}{N}\sum_{i=1}^N \delta_{Z_i^N(\cdot)} \in \mathcal{P}\big(C([0,S];\mathbb{R}^6)\big).$$

Then

$$\mathfrak{M}_{[0,S]}^N \xrightarrow[N\to\infty]{\mathbb{P}} \Pi_{[0,S]}^\infty \qquad \text{in } \mathcal{P}\big(C([0,S];\mathbb{R}^6)\big).$$

Equivalently, for every bounded continuous path functional $\Psi : C([0,S];\mathbb{R}^6) \to \mathbb{R}$,

$$\frac{1}{N}\sum_{i=1}^N \Psi\big(Z_i^N(\cdot)\big) \xrightarrow[N\to\infty]{\mathbb{P}} \int_{C([0,S];\mathbb{R}^6)} \Psi(\gamma)\,\Pi_{[0,S]}^\infty(d\gamma).$$

In particular, for every integer $0 \le t < S$,

$$M_t^N = (\mathrm{ev}_{t,t+1})_\# \mathfrak{M}_{[0,S]}^N \xrightarrow[N\to\infty]{\mathbb{P}} \Lambda_t^\infty.$$

Proof. Write $E := C([0,S];\mathbb{R}^6)$. Because the invariant law π_N is exchangeable and the N-particle dynamics are permutation-equivariant, the stationary path law $\widehat{P}_{[0,S]}^N$ on E^N is exchangeable. Applying the same finite de Finetti reduction used in the companion convergence paper, now on the Polish space E, yields a mixing measure $\widehat{Q}_N \in \mathcal{P}(\mathcal{P}(E))$ with canonical choice

$$\widehat{Q}_N = \mathrm{Law}(\mathfrak{M}_{[0,S]}^N),$$

whose fixed-k path marginals are asymptotically mixtures of product path laws.

The one-particle path marginals are tight on E. Stationarity gives the same uniform second-moment bounds at every time as in the companion convergence paper, and the underdamped

Langevin coefficients have at most linear growth with Brownian noise only in velocity. Standard Kolmogorov estimates therefore give tightness of $\mathrm{Law}(Z_1^N(\cdot))$ on E, hence tightness of $\{\widehat{Q}_N\}_{N\geq 2}$ on $\mathcal{P}(\mathcal{P}(E))$. Let $\widehat{Q}_\infty$ be any subsequential limit.

We identify the support of $\widehat{Q}_\infty$ by the nonlinear martingale problem. Fix $\phi \in C_c^\infty(\mathbb{R}^6)$, times $0 \leq u \leq v \leq S$, and a bounded continuous functional G depending only on the path segment on $[0,u]$. For a measure $\mu \in \mathcal{P}(\mathbb{R}^6)$, write

$$\mathcal{L}_\mu \phi(x,v) := v \cdot \nabla_x \phi(x,v) + \big(-\nabla U(x) - \gamma v + F[\mu](x,v)\big) \cdot \nabla_v \phi(x,v) + \frac{\sigma^2}{2}\Delta_v \phi(x,v).$$

For the finite-N system,

$$\phi(Z_i^N(v)) - \phi(Z_i^N(u)) - \int_u^v \Big[\mathcal{L}_{\mathfrak{M}^N_{[0,S],s}} \phi(Z_i^N(s)) + \big(F_i^N(s) - F[\mathfrak{M}^N_{[0,S],s}](Z_i^N(s))\big) \cdot \nabla_v \phi(Z_i^N(s))\Big]\, ds$$

is a martingale. Averaging over i and multiplying by $G(Z_i^N|_{[0,u]})$ gives zero expectation. Here

$$\mathfrak{M}^N_{[0,S],s} := (\mathrm{ev}_s)_\# \mathfrak{M}^N_{[0,S]} = \frac{1}{N}\sum_{j=1}^N \delta_{Z_j^N(s)}.$$

The force-defect estimate from the companion convergence paper, integrated over $s \in [u,v]$, shows that the second term inside the time integral contributes only $O(N^{-1/2})$. Passing to the limit and using continuity of $\mu \mapsto F[\mu]$ on moment-bounded classes, every path law η in the support of $\widehat{Q}_\infty$ satisfies

$$\int_E G(\gamma|_{[0,u]})\Big[\phi(\gamma(v)) - \phi(\gamma(u)) - \int_u^v \mathcal{L}_{\eta_s}\phi(\gamma(s))\, ds\Big]\eta(d\gamma) = 0,$$

where $\eta_s := (\mathrm{ev}_s)_\#\eta$. Thus $\widehat{Q}_\infty$-almost every η solves the McKean–Vlasov martingale problem on $[0,S]$.

Because each $\widehat{P}^N_{[0,S]}$ is stationary in time, every $\eta \in \operatorname{supp}\widehat{Q}_\infty$ is also stationary. Its time-0 marginal is the limit of the first marginals $\pi_{N,1}$ and therefore equals π_∞ by the stationary propagation-of-chaos theorem of the companion paper. The McKean–Vlasov SDE is well posed for every initial law with finite second moment, so there is a unique path law with initial law π_∞. Since π_∞ is stationary, that unique law is exactly $\Pi^\infty_{[0,S]}$. Hence $\widehat{Q}_\infty = \delta_{\Pi^\infty_{[0,S]}}$. Every subsequence has the same limit, so

$$\mathrm{Law}(\mathfrak{M}^N_{[0,S]}) \Rightarrow \delta_{\Pi^\infty_{[0,S]}},$$

which is equivalent to the stated convergence in probability. The final claim follows by the continuous mapping theorem for the evaluation map $\mathrm{ev}_{t,t+1}$. □

Theorem 9.8 (Stationary thermodynamic limit of plaquette observables)**.** *Let $\Phi : (\mathbb{R}^{12})^2 \to \mathbb{R}$ be bounded and continuous, and assume that $\chi_{\square,t}\Phi$ is continuous at $(\Lambda_t^\infty \otimes \Lambda_t^\infty)$-almost every pair. Then*

$$\int_{(\mathbb{R}^{12})^2} \Phi\, d\Gamma^N_{\square,t} \xrightarrow[N\to\infty]{\mathbb{P}} \iint_{(\mathbb{R}^{12})^2} \chi_{\square,t}(\zeta,\zeta')\,\Phi(\zeta,\zeta')\,\Lambda_t^\infty(d\zeta)\,\Lambda_t^\infty(d\zeta').$$

Proof. Set

$$\Psi_{\square,t}(\zeta,\zeta') := \chi_{\square,t}(\zeta,\zeta')\,\Phi(\zeta,\zeta').$$

By Definition 9.5,

$$\int \Phi\, d\Gamma^N_{\square,t} = \frac{1}{N^2}\sum_{i\neq j} \Psi_{\square,t}\big(\widetilde{Z}_i^N(t), \widetilde{Z}_j^N(t)\big)$$

$$= \iint \Psi_{\square,t}(\zeta,\zeta')\, M_t^N(d\zeta)\, M_t^N(d\zeta') \;-\; R^N_{\square,t},$$

where $|R^N_{\square,t}| \le \|\Psi_{\square,t}\|_\infty/N$. Hence $R^N_{\square,t} \to 0$ in probability. Because M_t^N converges in probability to the deterministic measure Λ_t^∞ by Theorem 9.7, the product measure $M_t^N \otimes M_t^N$ converges in law to $\Lambda_t^\infty \otimes \Lambda_t^\infty$. The boundedness and almost-everywhere continuity of $\Psi_{\square,t}$ then yield

$$\iint \Psi_{\square,t}(\zeta,\zeta')\, M_t^N(d\zeta)\, M_t^N(d\zeta') \xrightarrow[N\to\infty]{\mathbb{P}} \iint \Psi_{\square,t}(\zeta,\zeta')\, \Lambda_t^\infty(d\zeta)\, \Lambda_t^\infty(d\zeta').$$

Combining the last two displays proves the theorem. □

Remark 9.9. Theorems 9.2 and 9.8 are the thermodynamic statements needed for the gauge sector. The first passes link observables to the mean-field limit; the second does the same for the canonical plaquettes used by the Wilson action. No external discretization of the limiting law is introduced: the dynamic graph and its elementary 2-cells are carried directly to the mean-field regime by the same label-invariant graph rule. The companion convergence paper proves the one-time stationary propagation of chaos used in Theorem 9.2; the present Theorem 9.7 upgrades that control to stationary path space, and Theorem 9.8 is then the corresponding plaquette-level corollary.

Proposition 9.10 (Graph rule is not determined by particle paths)**.** *The stationary mean-field convergence results above do not determine the companion graph rule $\mathcal{P}_t^{(N)}$. More precisely, the plaquette geometry quantities*

$$\{A_P^{(N)}\}_{P\in\mathcal{C}_\square^{(N)}}, \qquad |\mathcal{C}_\square^{(N)}|, \qquad \sum_{P\in\mathcal{C}_\square^{(N)}} \Sigma_P^{(N),\mu\nu}\Sigma_P^{(N),\rho\sigma}\,\delta_{z_P^{(N)}}$$

depend on $\mathcal{P}_t^{(N)}$ and are not determined by the empirical particle path laws alone. Consequently, the mean-field plaquette measure $d\Gamma_\square^\infty$ of Definition 9.12 inherits its structure from the label-invariant pair rule χ_t, which must be specified as part of the model definition.

Proof. In Definition 3.1, the recorded companion sets $\{\mathcal{P}_t^{(N)}\}_{t\in\mathbb{T}}$ are part of the lattice input. Keeping the same particle trajectories fixed while changing the label-invariant companion rule changes the link set $\mathcal{E}^{(N)}$, the plaquette family $\mathcal{C}_\square^{(N)}$, and the associated area bivectors and plaquette counts. The empirical one-time and path measures of the particles are unaffected by such a change, hence the plaquette geometry is not determined by propagation of chaos alone. □

9.3 Mean-Field Yang-Mills Action

In the $N\to\infty$ thermodynamic regime the discrete interaction graph disappears and the relevant geometric objects are the stationary mean-field law π_∞ and its natural product measures. The bilocal gauge fields and the Wilson action are defined directly on the continuous probability space $(\operatorname{supp}\pi_\infty, \pi_\infty)$.

Definition 9.11 (Bilocal color connection)**.** *Let π_∞ be the stationary mean-field law from the companion convergence paper [1], and let $c : \operatorname{supp}\pi_\infty \to \mathbb{C}^3$ be the color field inherited from Definition 5.1. For π_∞-almost every pair (z,z'), define the* bilocal $\mathfrak{su}(3)$-valued connection

$$A(z,z') := \frac{1}{2}\sum_{a=1}^{8} \operatorname{Re}\!\big(c(z)^\dagger \lambda_a \big(c(z') - c(z)\big)\big)\lambda_a \;\in\; \mathfrak{su}(3).$$

The associated $SU(3)$ parallel transporter is

$$U(z,z') := \exp\bigl(ig_3 A(z,z')\bigr) \in SU(3).$$

For $\pi_\infty^{\otimes 4}$-almost every quadruple (z_1,z_2,z_3,z_4), the mean-field Wilson plaquette *is*

$$W(z_1,z_2,z_3,z_4) := U(z_1,z_2)\,U(z_2,z_3)\,U(z_3,z_4)\,U(z_4,z_1) \in SU(3).$$

Definition 9.12 (Mean-field plaquette measure)**.** *Let $\chi_\square : \operatorname{supp}\pi_\infty^{\otimes 4} \to \{0,1\}$ be the label-invariant quadruple indicator inherited from Definition 9.1 by the thermodynamic limit of Theorem 9.8. The* mean-field plaquette measure *is the positive measure on $\operatorname{supp}\pi_\infty^{\otimes 4}$ defined by*

$$d\Gamma_\square^\infty(z_1,z_2,z_3,z_4) := \chi_\square(z_1,z_2,z_3,z_4)\,\pi_\infty(dz_1)\,\pi_\infty(dz_2)\,\pi_\infty(dz_3)\,\pi_\infty(dz_4).$$

Theorem 9.13 (Mean-field Wilson action)**.** *The* mean-field Wilson action *is*

$$S_{\mathrm{W}}^\infty := \iiiint \Bigl(1 - \tfrac{1}{3}\operatorname{Re}\operatorname{tr} W(z_1,z_2,z_3,z_4)\Bigr)\, d\Gamma_\square^\infty(z_1,z_2,z_3,z_4).$$

By Theorem 9.8, in the $N \to \infty$ thermodynamic limit,

$$S_{\mathrm{W}}^{(N)} \xrightarrow[N\to\infty]{\mathbb{P}} S_{\mathrm{W}}^\infty.$$

The limit action S_{W}^∞ is gauge-invariant under simultaneous $SU(3)$ rotations $c(z) \mapsto \Omega\, c(z)$ for any fixed $\Omega \in SU(3)$, by the same argument as Proposition 6.9.

Proof. Apply Theorem 9.8 with $\Phi(z_1,z_2,z_3,z_4) := 1 - \frac{1}{3}\operatorname{Re}\operatorname{tr} W(z_1,z_2,z_3,z_4)$, which is bounded and continuous on $\operatorname{supp}\pi_\infty^{\otimes 4}$. The convergence in probability follows immediately. Gauge invariance is inherited from the finite-N Wilson action (Proposition 6.9) by passage to the limit. □

Remark 9.14 (Thermodynamic regularization). The mean-field action S_{W}^∞ is defined entirely within the probability space $(\operatorname{supp}\pi_\infty, \pi_\infty)$. Its regularization is provided by the viscous kernel length scale ℓ_{visc} and the smooth Gaussian kernel K_ρ. The asymptotic regime is the thermodynamic limit $N \to \infty$: the continuous stationary measure π_∞ *is* the spacetime geometry of the mean-field theory.

9.4 LSI-induced scalar witness sector

The gauge sector has a genuine thermodynamic limit on π_∞. For the scalar sector used in the quantization paper, we now make the same thermodynamic input explicit and fix a canonical free massive witness field.

Definition 9.15 (Mean-field scalar witness observable)**.** *Let π_∞ be the stationary mean-field law from the companion convergence paper [1]. Define the scalar witness observable by*

$$\sigma_\phi(z) := \|F[\pi_\infty](z)\|, \qquad z = (x,v) \in \mathbb{R}^6,$$

where $F[\pi_\infty](x,v) = \nu_{\mathrm{visc}}(A[\pi_\infty](x) - v)$ is the mean-field viscous drift from the convergence paper.

Proposition 9.16 (Thermodynamic control of the scalar witness)**.** *The observable σ_ϕ is globally Lipschitz on $\mathbb{R}^6$ and belongs to $L^2(\pi_\infty)$. More precisely, if L_{mf} is the finite constant supplied by the mean-field alignment-field Lipschitz estimate of [1], evaluated on the single-law class $\{\pi_\infty\}$, then*

$$|\sigma_\phi(z) - \sigma_\phi(z')| \le L_{\mathrm{mf}}\|z - z'\| \qquad \forall z, z' \in \mathbb{R}^6.$$

Proof. For any $z, z' \in \mathbb{R}^6$,

$$|\sigma_\phi(z) - \sigma_\phi(z')| \le \|F[\pi_\infty](z) - F[\pi_\infty](z')\|.$$

Applying the mean-field alignment-field Lipschitz estimate of [1] with $\mu = \nu = \pi_\infty$ gives the stated Lipschitz bound.

To prove square integrability, fix $z_0 = (0, 0) \in \mathbb{R}^6$. The Lipschitz estimate implies

$$\sigma_\phi(z) \le \sigma_\phi(z_0) + L_{\mathrm{mf}}\|z - z_0\| \le \sigma_\phi(z_0) + L_{\mathrm{mf}}\|z\|.$$

The stationary propagation-of-chaos theorem of [1], together with the uniform second-moment estimate proved there for the stationary marginals, implies that π_∞ has finite second moment on $\mathbb{R}^6$. Therefore $\sigma_\phi \in L^2(\pi_\infty)$. $\square$

Definition 9.17 (Scalar witness mass)**.** *Let $C_{\mathrm{LSI}} < \infty$ be the mean-field logarithmic Sobolev constant of the companion convergence paper [1], and let L_{mf} be as in Proposition 9.16. Define the scalar witness mass by*

$$m_\phi := \left(1 + C_{\mathrm{LSI}} L_{\mathrm{mf}}^2\right)^{-1/2} > 0.$$

Remark 9.18 (Interpretation of the scalar witness sector). The scalar matter field in Definition 7.1 is therefore not an arbitrary local-potential sector. It is the canonical free massive scalar witness attached to the thermodynamic stiffness scale already certified by the mean-field LSI on π_∞. No additional scalar local potential is introduced anywhere in the series.

9.5 Quantization and partition function

The data package $\mathcal{Q}$ provides a well-defined action $S = S_{\mathrm{W}} + S_{\mathrm{F}} + S_\phi$ on the finite-dimensional field variables $\{\psi_x, \bar{\psi}_x, \phi_x\}_{x \in \mathcal{N}}$. The path integral over these fields,

$$Z = \int \Big[\prod_{x \in \mathcal{N}} d\psi_x \, d\bar{\psi}_x \, d\phi_x\Big] \exp\!\big(-S_{\mathrm{W}} - S_{\mathrm{F}} - S_\phi\big),$$

is a finite-dimensional ordinary integral over finite-dimensional fermionic Grassmann variables and the scalar sector $\mathbb{R}^{|\mathcal{N}|}$, evaluated in the background of the explicit finite link configuration $\{U_3(e)\}_{e \in \mathcal{E}}$. Because the scalar action is free massive with $m_\phi > 0$, the scalar sector is already Gaussian and therefore finite before any further gauge promotion. If one wishes to promote the link variables themselves to dynamical integration variables, gauge-fixing via Faddeev–Popov is straightforward in the finite setting. The resulting Z is well-defined without any ultraviolet regularization; all divergences that would appear in a continuum QFT are absent because the lattice is finite. Evaluating Z numerically for realistic run data and comparing with continuum expectations is a natural next step.

10 Conclusions

We have constructed a complete, finite, and $SU(3)$ gauge-invariant lattice QFT data package $\mathcal{Q}$ from the discrete trajectory data of a viscous multi-particle system. The construction proceeds in six conceptual steps:

1. Build the spacetime lattice $(\mathcal{N}, \mathcal{E})$ from the recorded sites, worldline links, and same-time interaction links.

2. Compute the row-normalized viscous force $\mathbf{F}^{\mathrm{visc}}(x)$ at each site from the interaction-link weights.

3. Form complex color vectors $\mathbf{c}_x \in \mathbb{C}^3$ by modulating each force component with a de Broglie kinematic phase and normalizing.

4. Define the $SU(3)$ link transport $U_3(e) = \exp(i\Theta_e)$ by projecting the short-link color increment $\mathbf{c}_x^\dagger \lambda_a(\mathbf{c}_y - \mathbf{c}_x)$ onto the Lie generator space $\mathfrak{h}_3$ via the Gell-Mann basis.

5. Build the canonical plaquette family $\mathcal{C}_\square$, the area bivectors Σ_P, the discrete plaquette holonomies $U_3(P)$, and the Wilson action S_{W}.

6. Prove the $N \to \infty$ thermodynamic mean-field limit for both link and plaquette observables, and define the bilocal mean-field Yang–Mills action S_{W}^∞ directly on the stationary measure π_∞.

All objects are explicit and finite (Theorem 8.1). The gauge group $SU(3)$ is not imposed but emerges from the three-dimensional viscous dynamics together with the Gell-Mann projection. In the stationary thermodynamic regime, the same interaction graph has a deterministic mean-field limit for link observables in the sense of Theorem 9.2, and the canonical plaquettes have the analogous deterministic mean-field limit by Theorems 9.7 and 9.8. In the thermodynamic regime, Theorem 9.13 identifies the limit of the discrete Wilson action as the bilocal mean-field action S_{W}^∞ on the stationary measure π_∞, with regularization provided by the viscous kernel K_ρ.

Two major extensions are deferred to companion papers: (i) path-integral quantization of the mean-field theory with the bilocal action S_{W}^∞ and the Markov-chain measure on the interaction graph; (ii) axiom verification — checking that $\mathcal{Q}$ satisfies the Osterwalder–Schrader axioms evaluated on the integral operator of the continuous mean-field law.

References

[1] Convergence, hypocoercivity, and mean-field limit for the kinetic core. Manuscript, 2026. Source file: `docs/source/4_ymmg/01_convergence.tex`.